

Jeffrey Hu

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EDUCATION

University of Cambridge

PhD student

Current

Cambridge, UK

Supervisor: Professor Ayush Tewari. Research in 3D computer vision, computer graphics, differentiable rendering, and generative models of the real world.

Massachusetts Institute of Technology

Computer Science and Engineering

2018, 2019

Cambridge, MA

Master of Science, 2019; Bachelor of Science, 2018. GPA: 4.7/5.0.

Research advisor: Professor Antonio Torralba.

EXPERIENCE

Inria — GraphDeco

Research Engineer

Previously

- Worked with Professor George Drettakis on 3D Gaussian splatting and differentiable rendering, including intrinsic decomposition, editable reflections, and reconstruction of scenes with oscillatory motion.

gsplat

Open-source Contributor

From Apr 2024

- Contributed to an open-source Gaussian splatting library for 3D reconstruction and novel-view synthesis, including:
 - [3DGS-MCMC](#) for improved densification and [bilateral grids](#) for handling exposure changes.
 - [Fisheye camera support](#) and [2.5D Gaussian splatting](#) for surface reconstruction.
 - Combined MCMC, Self-Organizing Gaussian Splats, and spherical-harmonics k-means clustering for [Gaussian splat compression](#).
 - [Camera defocus blur modeling](#).

Parallel Systems

Senior Software Engineer, Perception

Mar 2022–Apr 2024

Los Angeles, CA

- Led development of a real-time multitask computer vision network and data engine for an autonomous train perception system.
- Iteratively built a multitask network for 3D object detection, 3D track prediction, depth prediction, and bird's-eye-view segmentation.
 - Wrote a PyTorch library for training models with various backbones, task-specific heads, and loss functions.
 - Developed a representation for railroad track geometry by regressing 3D splines and classifying switch states.
 - Shipped multiple model iterations onto a Jetson using Rust, TensorRT, and DeepStream.
- Developed a 3D reconstruction and autolabeling pipeline for scalable ground-truth generation using Dagster and Kubernetes.
 - Used zero-shot, open-vocabulary foundation models for object detection, segmentation, tracking, and monocular depth estimation.
 - Aggregated predictions across time to reconstruct 3D scene representations as high-quality labels for training and evaluating the real-time network.
 - Built infrastructure to balance datasets, mine scenarios, and automatically generate test reports to support the regulatory safety case over time.
- Contributed to sensor calibration, onboard data collection, triggering, and upload infrastructure.

EXPERIENCE, CONTINUED**TuSimple***Research Engineer II*

Jan 2020–Mar 2022

San Diego, CA

- Developed a method for correcting camera drift in visual odometry by training a network to regress the difference between the current image and a rendered synthetic map view.
- Wrote an onboard C++ library to efficiently manage 6DoF transformation matrices between coordinate frames of a scene graph in an asynchronous system.

MIT Computer Vision Group*Research Assistant*

Sep 2017–Sep 2019

Cambridge, MA

- Investigated scalable semi-automatic methods for labeling large segmentation datasets.
- Built a browser-based annotation tool for rapid segmentation editing.

Amazon Prime Air*Software Engineer Intern*

Jun 2016–Aug 2016

Seattle, WA

PUBLICATIONS**Intrinsic Decomposition and Editing of 3D Gaussian Splats**A. Lanvin, **J. Hu**, S. Lucas, A. Bousseau, and G. Drettakis*ISD / Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 2026.**Adaptive Spatio-Temporal 3D Gaussian Splatting for Scenes with Oscillatory Motion**P. Tzathas, **J. Hu**, A. Meuleman, G. Cordonnier, and G. Drettakis*Eurographics / Computer Graphics Forum*, 2026.**Efficient Camera-Controlled Video Generation of Static Scenes via Sparse Diffusion and 3D Rendering**J. Chen, **J. Hu**, J. Lasenby, and A. Tewari*arXiv preprint*, 2026.**Editable Physically-based Reflections in Raytraced Gaussian Radiance Fields**Y. Poirier-Ginter, **J. Hu**, J.-F. Lalonde, and G. Drettakis*SIGGRAPH Asia*, 2025.**gsplat: An Open-Source Library for Gaussian Splatting**V. Ye, R. Li, J. Kerr, M. Turkulainen, B. Yi, Z. Pan, O. Seiskari, J. Ye, **J. Hu**, M. Tancik, and A. Kanazawa*Journal of Machine Learning Research*, 2025.LEADERSHIP & INTERESTS**MIT Varsity Squash***Captain*

Sep 2014–Jun 2018

Cambridge, MA

Programming: Python, PyTorch, C++, Rust, CUDA, Git.**Interests:** Squash, cycling, skiing, hiking.